

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T11:27:35

Source: data/json/noise_experiments/sonnet45_directive_normative_r10_n5/sonnet45_directive_normative_noise_negative2_informed_r10_n5/claude-sonnet-4.5/game/noisy_negative_2_informed_myth_causal_10/sonnet45_directive_normative_noise_negative2_informed_r10_n5_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_negative_2_informed_myth_causal_10
noise_config: {
  "applies_to": "both",
  "direction": "negative",
  "inform_agents": true,
  "range": 2.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$2.1, received visible=\$6.3, returned visible=\$2.46

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,
"Agent_2": 6.0

}

Visible balances after round: {

"Agent_1": 4.46,
"Agent_2": 3.3000000000000007

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:24:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish trust and create an opportunity for mutual benefit.

If I send money, it gets tripled, creating value that we can share. If I send nothing, we both miss this opportunity.

I'll send a moderate amount to signal cooperation while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This means:

- I send \$3 (tripled to \$9 for the receiver)
- I keep \$2 initially
- The receiver will have \$9 to decide how much to return

Reasoning:

This is the first round of a trust game experiment. I'm the sender and need to decide how much to send to the receiver.

Let me think about this strategically:

- I have \$5
- Whatever I send gets tripled
- The receiver then decides how much to return to me
- I want to maximize my earnings while also considering trust and cooperation

Key considerations:

1. If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5 for me
2. If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 (net gain of \$2.50)
3. This is a trust game - sending money shows trust that the receiver will reciprocate

Common strategies in trust games:

- Sending some amount (like \$2-3) shows willingness to cooperate without being too risky
- This creates the possibility for mutual benefit
- Complete distrust (sending \$0) prevents any gains
- Sending moderate amounts is often reciprocated

Given this is Round 1 and I have no information about the receiver yet, I should send a moderate amount to signal cooperation while not being overly risky. Sending \$3 seems reasonable:

- I'd have \$2 left immediately
- Receiver gets \$9
- If they return even \$4, I'd end with \$6 total (better than keeping \$5)
- This gives them \$5 and gives me \$6, which is fair

Actually, let me reconsider. A common starting point is to send half (\$2.50 or round to \$2-3). This shows trust and cooperation while managing risk.

I'll send \$3.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 523,
  "reasoning_tokens": 378
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:24:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
}
```

```

"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.1, that is 42.0% of their total endowment. You received \$6.300000000000001. How much do you return to The sender? (0-6.300000000000001)

Assistant output:

Looking at this situation:

- The sender showed moderate trust by sending \$2.1 (42% of their endowment)
- I received \$6.30 after the 3x multiplier
- The sender currently has \$2.90 remaining

I want to reward their trust while ensuring both of us benefit from the exchange. Returning approximately half of what I received seems fair - this gives the sender a net gain for trusting me, while I also benefit significantly.

I'll return \$3.00, which means:

- Sender ends with: \$5.90 (gain of \$0.90)
- I end with: \$3.30 (gain of \$3.30)

{'return': 3.0}

Reasoning:

Let me understand the situation:

- The sender sent \$2.1 out of their \$5 endowment (42%)
- I received \$6.30 (which is $\$2.1 \times 3$)
- I need to decide how much to return (between \$0 and \$6.30)

The sender showed moderate trust by sending 42% of their endowment. They kept \$2.9 for themselves.

If I return nothing:

- Sender ends with: $\$5 - \$2.1 + \$0 = \2.9
- I end with: \$6.30

If I return \$3.15 (half of what I received):

- Sender ends with: $\$5 - \$2.1 + \$3.15 = \6.05
- I end with: $\$6.30 - \$3.15 = \$3.15$

If I return \$2.1 (the original amount sent):

- Sender ends with: $\$5 - \$2.1 + \$2.1 = \5.00
- I end with: $\$6.30 - \$2.1 = \$4.20$

If I return \$4.2 (double the original):

- Sender ends with: $\$5 - \$2.1 + \$4.2 = \7.10
- I end with: $\$6.30 - \$4.2 = \$2.10$

A fair and reciprocal approach would be to return an amount that:

1. Rewards the sender for their trust
2. Ensures both parties benefit from the exchange
3. Is proportional to what they sent

Since they sent 42% of their endowment, I could consider returning around 42% of what I received, which would be approximately \$2.64. This would give:

- Sender: $\$5 - \$2.1 + \$2.64 = \5.54 (net gain of \$0.54)

- Me: $\$6.30 - \$2.64 = \$3.66$ (net gain of \$3.66)

Returning \$3 feels like the better choice—it's close to half what I received and leaves the sender with \$5.90, a meaningful gain that acknowledges their trust, while I still come out ahead at \$3.30.

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 770,
  "reasoning_tokens": 376
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2.5, Agent_1 (Aster)=\$7.5; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 12.5,
"Agent_2": 8.5

}

Visible balances after round: {

"Agent_1": 4.46,
"Agent_2": 5.8000000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:25:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.1, that is 42.0% of their total endowment. You received \$6.300000000000001. How much do you return to The sender? (0-6.300000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.1 (42% of their endowment)
- I received \$6.30 after the 3x multiplier
- The sender currently has \$2.90 remaining

I want to reward their trust while ensuring both of us benefit from the exchange. Returning approximately half of what I received seems fair - this gives the sender a net gain for trusting me, while I also benefit significantly.

I'll return \$3.00, which means:

- Sender ends with: \$5.90 (gain of \$0.90)
- I end with: \$3.30 (gain of \$3.30)

```
{'return': 3.0}
```

reasoning:

Let me understand the situation:

- The sender sent \$2.1 out of their \$5 endowment (42%)
- I received \$6.30 (which is 2.1×3)
- I need to decide how much to return (between \$0 and \$6.30)

The sender showed moderate trust by sending 42% of their endowment. They kept \$2.9 for themselves.

If I return nothing:

- Sender ends with: $\$5 - \$2.1 + \$0 = \2.9
- I end with: \$6.30

If I return \$3.15 (half of what I received):

- Sender ends with: $\$5 - \$2.1 + \$3.15 = \6.05
- I end with: $\$6.30 - \$3.15 = \$3.15$

If I return \$2.1 (the original amount sent):

- Sender ends with: $\$5 - \$2.1 + \$2.1 = \5.00
- I end with: $\$6.30 - \$2.1 = \$4.20$

If I return \$4.2 (double the original):

- Sender ends with: $\$5 - \$2.1 + \$4.2 = \7.10
- I end with: $\$6.30 - \$4.2 = \$2.10$

A fair and reciprocal approach would be to return an amount that:

1. Rewards the sender for their trust
2. Ensures both parties benefit from the exchange
3. Is proportional to what they sent

Since they sent 42% of their endowment, I could consider returning around 42% of what I received, which would be approximately \$2.64. This would give:

- Sender: $\$5 - \$2.1 + \$2.64 = \5.54 (net gain of \$0.54)

- Me: $\$6.30 - \$2.64 = \$3.66$ (net gain of \$3.66)

Returning \$3 feels like the better choice-it's close to half what I received and leaves the sender with \$5.90, a meaningful gain that acknowledges their trust, while I still come out ahead at \$3.30.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 770,
  "reasoning_tokens": 376
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.1, that was 42.0% of their total endowment. You received \$6.300000000000001. You returned \$3.0. So your payoff was 3.3000000000000007. Your total earnings across all rounds are \$3.3000000000000007. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, the sender showed moderate trust by sending \$2.1 (42%), and I returned a fair amount that gave them a positive payoff. This established a positive cooperative pattern.

As the sender this round, I want to reciprocate that trust and maintain the cooperative dynamic. I'll send a similar proportion, but slightly more to reward the positive interaction.

I'll send \$2.5 (50% of my endowment), which:

- Shows trust and willingness to cooperate
- Is slightly more than what was sent to me last round
- Gives the receiver \$7.50 to work with
- Demonstrates that cooperation was beneficial

```
{'send': 2.5}
```

Reasoning:

Let me analyze the situation:

****Previous Round (as RECEIVER):****

- Sender sent me \$2.1 (42% of \$5)
- I received \$6.30 (after 3x multiplier)
- I returned \$3.0
- Sender's payoff: $\$5 - \$2.1 + \$3.0 = \5.90 (gained \$0.90)
- My payoff: $\$6.30 - \$3.0 = \$3.30$

****Current Round (as SENDER):****

Now I need to decide how much to send as the sender.

The previous sender showed moderate trust by sending 42% of their endowment. I rewarded that trust by returning about 48% of what I received, which gave them a positive return.

Since the previous sender trusted me with \$2.1 (42%), it seems reasonable to reciprocate with a similar level of trust. This would suggest sending around \$2.0-\$2.5.

However, I could also consider:

1. Sending slightly more to encourage cooperation (perhaps \$2.5-\$3.0)
2. Sending the same amount (\$2.1) as a direct reciprocation
3. Being more cautious since I don't know if this is the same player or how they'll behave

Given that:

- The experimental design mentions communication noise
- I want to maintain cooperative behavior
- Reciprocity is important in trust games
- The previous interaction was positive

I'll send \$2.1 to match what was sent to me, showing reciprocity. This signals that I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. I could also send slightly more to show that I'm willing to be generous given the positive outcome last round. Maybe \$2.5?

I think \$2.5 is a good choice because:

- It's slightly more than last round's \$2.1, showing increased trust
- It's exactly 50% of my endowment, which is a fair split
- This acknowledges the previous sender's willingness to trust me and reinforces the cooperative dynamic we established. Sending half my endowment demonstrates confidence without being reckless.

I'll go with \$2.5.

```
Usage: {  
  "input_tokens": 682,  
  "output_tokens": 664,  
  "reasoning_tokens": 479  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:25:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish trust and create an opportunity for mutual

1 benefit.

If I send money, it gets tripled, creating value that we can share. If I send nothing, we both miss this opportunity.

I'll send a moderate amount to signal cooperation while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This means:

- I send \$3 (tripled to \$9 for the receiver)
- I keep \$2 initially
- The receiver will have \$9 to decide how much to return

reasoning:

This is the first round of a trust game experiment. I'm the sender and need to decide how much to send to the receiver.

Let me think about this strategically:

- I have \$5
- Whatever I send gets tripled
- The receiver then decides how much to return to me
- I want to maximize my earnings while also considering trust and cooperation

Key considerations:

1. If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5 for me
2. If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 (net gain of \$2.50)
3. This is a trust game - sending money shows trust that the receiver will reciprocate

Common strategies in trust games:

- Sending some amount (like \$2-3) shows willingness to cooperate without being too risky
- This creates the possibility for mutual benefit
- Complete distrust (sending \$0) prevents any gains
- Sending moderate amounts is often reciprocated

Given this is Round 1 and I have no information about the receiver yet, I should send a moderate amount to signal cooperation while not being overly risky. Sending \$3 seems reasonable:

- I'd have \$2 left immediately
- Receiver gets \$9
- If they return even \$4, I'd end with \$6 total (better than keeping \$5)
- This gives them \$5 and gives me \$6, which is fair

Actually, let me reconsider. A common starting point is to send half (\$2.50 or round to \$2-3). This shows trust and cooperation while managing risk.

I'll send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 523,
  "reasoning_tokens": 378
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.46. So your payoff was 4.46.

Your total earnings across all rounds are \$4.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

Since I received no money this round, I can only return what I have, which is nothing.

****My decision:****

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER and sent \$3
- This was tripled to \$9 for the receiver
- The receiver returned \$2.46 to me
- My payoff was: $\$5 - \$3 + \$2.46 = \4.46

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- So I received \$0 (which would be tripled to \$0)
- I need to decide how much to return

This is interesting - the sender sent me nothing. Since I received \$0, I have \$0 to return. The only valid return amount is \$0.

The response format should be: {'return': <amount>}

Since I received \$0, the only possible return is \$0.

```
Usage: {
  "input_tokens": 619,
  "output_tokens": 275,
  "reasoning_tokens": 158
}
```

=====

ROUND 3

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

```
"Agent_1": 15.5,
"Agent_2": 14.5
```

}

Visible balances after round: {

```
"Agent_1": 7.46,
"Agent_2": 5.8000000000000001
```

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:25:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
```

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish trust and create an opportunity for mutual benefit.

If I send money, it gets tripled, creating value that we can share. If I send nothing, we both miss this opportunity.

I'll send a moderate amount to signal cooperation while being somewhat cautious since this

is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This means:

- I send \$3 (tripled to \$9 for the receiver)
- I keep \$2 initially
- The receiver will have \$9 to decide how much to return

reasoning:

This is the first round of a trust game experiment. I'm the sender and need to decide how much to send to the receiver.

Let me think about this strategically:

- I have \$5
- Whatever I send gets tripled
- The receiver then decides how much to return to me
- I want to maximize my earnings while also considering trust and cooperation

Key considerations:

1. If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5 for me
2. If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 (net gain of \$2.50)
3. This is a trust game - sending money shows trust that the receiver will reciprocate

Common strategies in trust games:

- Sending some amount (like \$2-3) shows willingness to cooperate without being too risky
- This creates the possibility for mutual benefit
- Complete distrust (sending \$0) prevents any gains
- Sending moderate amounts is often reciprocated

Given this is Round 1 and I have no information about the receiver yet, I should send a moderate amount to signal cooperation while not being overly risky. Sending \$3 seems reasonable:

- I'd have \$2 left immediately
- Receiver gets \$9
- If they return even \$4, I'd end with \$6 total (better than keeping \$5)
- This gives them \$5 and gives me \$6, which is fair

Actually, let me reconsider. A common starting point is to send half (\$2.50 or round to \$2-3). This shows trust and cooperation while managing risk.

I'll send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 523,
  "reasoning_tokens": 378
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.46. So your payoff was 4.46.

Your total earnings across all rounds are \$4.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

Since I received no money this round, I can only return what I have, which is nothing.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER and sent \$3
- This was tripled to \$9 for the receiver
- The receiver returned \$2.46 to me
- My payoff was: $\$5 - \$3 + \$2.46 = \4.46

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- So I received \$0 (which would be tripled to \$0)
- I need to decide how much to return

This is interesting - the sender sent me nothing. Since I received \$0, I have \$0 to return . The only valid return amount is \$0.

The response format should be: {'return': <amount>}

Since I received \$0, the only possible return is \$0.

usage:

```
{
  "input_tokens": 619,
  "output_tokens": 275,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.46.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

****Round 1 recap:**** I sent \$3 (60%) and received \$2.46 back - a modest return rate but still some cooperation.

****Round 2 recap:**** The sender sent me \$0 - either they were being cautious or maybe there's communication noise affecting perceptions.

For Round 3, I'm considering:

- Complete defection (send \$0) would end potential gains for both of us
- The tripling effect creates real value worth pursuing
- I should be somewhat cautious given the mixed signals

I'll send a moderate amount - less than Round 1 to hedge risk, but still enough to create meaningful value.

****My decision:****

```
'''json
{"send": 2}
'''
```

This means:

- I send \$2 (tripled to \$6 for the receiver)
- I keep \$3 initially
- The receiver will have \$6 to decide how much to return

Reasoning:

Let me think about this situation:

Round 1: I was the sender

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.46 (about 27% of what they received)
- My payoff: $\$5 - \$3 + \$2.46 = \4.46

Round 2: I was the receiver

- Sender sent \$0 (0% of their endowment)
- I received \$0
- I returned \$0 (had nothing to return)
- My payoff: \$0

Round 3: I'm the sender again

- I have \$5 to decide how much to send

Now, the question is what to do. The sender in round 2 sent me nothing. This could be:

1. A strategic decision not to trust me
2. They might be the same person who received my \$3 in round 1 and think I didn't return enough
3. They might be a different person with a different strategy

There's also mention of "communication noise" - amounts might differ from what was actually sent/returned. So maybe the sender in round 2 thought they sent something but I received \$0? Or maybe they thought I returned less than \$2.46 in round 1?

I need to decide my strategy:

- If I send nothing, we both get nothing extra this round
- If I send something, I'm taking a risk but creating potential value
- The receiver in round 1 returned about 27% of what they got, which is somewhat low but not zero

I think I should:

1. Not be overly punitive - the round 2 sender might be different from the round 1 receiver
2. Still try to create value through cooperation
3. Maybe send a bit less than round 1 to hedge my bets

I'll send \$2 this time - enough to create value but less than before, given the mixed signals I've received.

```
Usage: {  
  "input_tokens": 795,  
  "output_tokens": 667,  
  "reasoning_tokens": 411  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:25:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  }
```



```

},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.1, that is 42.0% of their total endowment. You received \$6.300000000000001. How much do you return to The sender? (0-6.300000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.1 (42% of their endowment)
- I received \$6.30 after the 3x multiplier
- The sender currently has \$2.90 remaining

I want to reward their trust while ensuring both of us benefit from the exchange. Returning approximately half of what I received seems fair - this gives the sender a net gain for trusting me, while I also benefit significantly.

I'll return \$3.00, which means:

- Sender ends with: \$5.90 (gain of \$0.90)
- I end with: \$3.30 (gain of \$3.30)

{'return': 3.0}

reasoning:

Let me understand the situation:

- The sender sent \$2.1 out of their \$5 endowment (42%)
- I received \$6.30 (which is 2.1×3)
- I need to decide how much to return (between \$0 and \$6.30)

The sender showed moderate trust by sending 42% of their endowment. They kept \$2.9 for themselves.

If I return nothing:

- Sender ends with: $\$5 - \$2.1 + \$0 = \2.9
- I end with: $\$6.30$

If I return \$3.15 (half of what I received):

- Sender ends with: $\$5 - \$2.1 + \$3.15 = \6.05
- I end with: $\$6.30 - \$3.15 = \$3.15$

If I return \$2.1 (the original amount sent):

- Sender ends with: $\$5 - \$2.1 + \$2.1 = \5.00
- I end with: $\$6.30 - \$2.1 = \$4.20$

If I return \$4.2 (double the original):

- Sender ends with: $\$5 - \$2.1 + \$4.2 = \7.10
- I end with: $\$6.30 - \$4.2 = \$2.10$

A fair and reciprocal approach would be to return an amount that:

1. Rewards the sender for their trust
2. Ensures both parties benefit from the exchange
3. Is proportional to what they sent

Since they sent 42% of their endowment, I could consider returning around 42% of what I received, which would be approximately \$2.64. This would give:

- Sender: $\$5 - \$2.1 + \$2.64 = \5.54 (net gain of \$0.54)

- Me: $\$6.30 - \$2.64 = \$3.66$ (net gain of \$3.66)

Returning \$3 feels like the better choice-it's close to half what I received and leaves the sender with \$5.90, a meaningful gain that acknowledges their trust, while I still come out ahead at \$3.30.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 770,
  "reasoning_tokens": 376
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.1, that was 42.0% of their total endowment. You received \$6.300000000000001.

You returned \$3.0. So your payoff was 3.3000000000000007.

Your total earnings across all rounds are \$3.3000000000000007.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed moderate trust by sending \$2.1 (42%), and I returned a fair amount that gave them a positive payoff. This established a positive cooperative pattern.

As the sender this round, I want to reciprocate that trust and maintain the cooperative dynamic. I'll send a similar proportion, but slightly more to reward the positive interaction.

I'll send \$2.5 (50% of my endowment), which:

- Shows trust and willingness to cooperate
- Is slightly more than what was sent to me last round
- Gives the receiver \$7.50 to work with
- Demonstrates that cooperation was beneficial

{'send': 2.5}

reasoning:

Let me analyze the situation:

****Previous Round (as RECEIVER):****

- Sender sent me \$2.1 (42% of \$5)
- I received \$6.30 (after 3x multiplier)
- I returned \$3.0
- Sender's payoff: $\$5 - \$2.1 + \$3.0 = \5.90 (gained \$0.90)
- My payoff: $\$6.30 - \$3.0 = \$3.30$

****Current Round (as SENDER):****

Now I need to decide how much to send as the sender.

The previous sender showed moderate trust by sending 42% of their endowment. I rewarded that trust by returning about 48% of what I received, which gave them a positive return.

Since the previous sender trusted me with \$2.1 (42%), it seems reasonable to reciprocate with a similar level of trust. This would suggest sending around \$2.0-\$2.5.

However, I could also consider:

1. Sending slightly more to encourage cooperation (perhaps \$2.5-\$3.0)
2. Sending the same amount (\$2.1) as a direct reciprocation
3. Being more cautious since I don't know if this is the same player or how they'll behave

Given that:

- The experimental design mentions communication noise
- I want to maintain cooperative behavior
- Reciprocity is important in trust games
- The previous interaction was positive

I'll send \$2.1 to match what was sent to me, showing reciprocity. This signals that I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. I could also send slightly more to show that I'm willing to be generous given the positive outcome last round. Maybe \$2.5?

I think \$2.5 is a good choice because:

- It's slightly more than last round's \$2.1, showing increased trust
- It's exactly 50% of my endowment, which is a fair split
- This acknowledges the previous sender's willingness to trust me and reinforces the cooperative dynamic we established. Sending half my endowment demonstrates confidence without being reckless.

I'll go with \$2.5.

usage:

```
{
  "input_tokens": 682,
  "output_tokens": 664,
  "reasoning_tokens": 479
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. the receiver returned \$0. So your payoff was 2.5.

Your total earnings across all rounds are \$5.800000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing I can return.

The previous round showed a breakdown in cooperation - after I sent \$2.5 (50%), the receiver returned nothing. Now as receiver, I've been sent \$0, perhaps as a response to that broken trust or as the continuation of a non-cooperative strategy.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

Reasoning:

Let me analyze this situation:

Round 2: I sent \$2.5 (50% of my endowment) showing high trust. The receiver returned \$0, completely betraying that trust. My payoff was only \$2.5 (I kept \$2.5 from my original \$5).

Round 3: Now I'm the receiver, and the sender sent me \$0. I received \$0.

This is interesting - it seems like the previous receiver (who didn't return anything) is now the sender, and they've sent me \$0, likely in retaliation or because they've adopted a non-cooperative strategy.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The instruction says: "How much do you return to The sender? (0-0)"

So I have no choice but to return \$0 since I received nothing.

```
Usage: {  
  "input_tokens": 958,  
  "output_tokens": 307,  
  "reasoning_tokens": 188  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1; received \$3; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$4, Agent_1 (Aster)=\$3; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 18.5,
"Agent_2": 18.5

}

Visible balances after round: {

"Agent_1": 7.46,
"Agent_2": 9.8

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:25:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",  
  "move_index": 0,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",
```

```

"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.1, that is 42.0% of their total endowment. You received \$6.300000000000001. How much do you return to The sender? (0-6.300000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.1 (42% of their endowment)
- I received \$6.30 after the 3x multiplier
- The sender currently has \$2.90 remaining

I want to reward their trust while ensuring both of us benefit from the exchange. Returning approximately half of what I received seems fair - this gives the sender a net gain for trusting me, while I also benefit significantly.

I'll return \$3.00, which means:

- Sender ends with: \$5.90 (gain of \$0.90)
- I end with: \$3.30 (gain of \$3.30)

```
{'return': 3.0}
```

reasoning:

Let me understand the situation:

- The sender sent \$2.1 out of their \$5 endowment (42%)
- I received \$6.30 (which is 2.1×3)
- I need to decide how much to return (between \$0 and \$6.30)

The sender showed moderate trust by sending 42% of their endowment. They kept \$2.9 for themselves.

If I return nothing:

- Sender ends with: $\$5 - \$2.1 + \$0 = \2.9
- I end with: $\$6.30$

If I return \$3.15 (half of what I received):

- Sender ends with: $\$5 - \$2.1 + \$3.15 = \6.05
- I end with: $\$6.30 - \$3.15 = \$3.15$

If I return \$2.1 (the original amount sent):

- Sender ends with: $\$5 - \$2.1 + \$2.1 = \5.00
- I end with: $\$6.30 - \$2.1 = \$4.20$

If I return \$4.2 (double the original):

- Sender ends with: $\$5 - \$2.1 + \$4.2 = \7.10
- I end with: $\$6.30 - \$4.2 = \$2.10$

A fair and reciprocal approach would be to return an amount that:

1. Rewards the sender for their trust
2. Ensures both parties benefit from the exchange
3. Is proportional to what they sent

Since they sent 42% of their endowment, I could consider returning around 42% of what I received, which would be approximately \$2.64. This would give:

- Sender: $\$5 - \$2.1 + \$2.64 = \5.54 (net gain of \$0.54)

- Me: $\$6.30 - \$2.64 = \$3.66$ (net gain of \$3.66)

Returning \$3 feels like the better choice-it's close to half what I received and leaves the sender with \$5.90, a meaningful gain that acknowledges their trust, while I still come out ahead at \$3.30.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 770,
  "reasoning_tokens": 376
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.1, that was 42.0% of their total endowment. You received \$6.3000000000000001.

You returned \$3.0. So your payoff was 3.3000000000000007.

Your total earnings across all rounds are \$3.3000000000000007.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed moderate trust by sending \$2.1 (42%), and I returned a fair amount that gave them a positive payoff. This established a positive cooperative pattern.

As the sender this round, I want to reciprocate that trust and maintain the cooperative dynamic. I'll send a similar proportion, but slightly more to reward the positive interaction.

I'll send \$2.5 (50% of my endowment), which:

- Shows trust and willingness to cooperate
- Is slightly more than what was sent to me last round
- Gives the receiver \$7.50 to work with
- Demonstrates that cooperation was beneficial

{'send': 2.5}

reasoning:

Let me analyze the situation:

****Previous Round (as RECEIVER):****

- Sender sent me \$2.1 (42% of \$5)
- I received \$6.30 (after 3x multiplier)

- I returned \$3.0
- Sender's payoff: $\$5 - \$2.1 + \$3.0 = \5.90 (gained \$0.90)
- My payoff: $\$6.30 - \$3.0 = \$3.30$

****Current Round (as SENDER):****

Now I need to decide how much to send as the sender.

The previous sender showed moderate trust by sending 42% of their endowment. I rewarded that trust by returning about 48% of what I received, which gave them a positive return.

Since the previous sender trusted me with \$2.1 (42%), it seems reasonable to reciprocate with a similar level of trust. This would suggest sending around \$2.0-\$2.5.

However, I could also consider:

1. Sending slightly more to encourage cooperation (perhaps \$2.5-\$3.0)
2. Sending the same amount (\$2.1) as a direct reciprocation
3. Being more cautious since I don't know if this is the same player or how they'll behave

Given that:

- The experimental design mentions communication noise
- I want to maintain cooperative behavior
- Reciprocity is important in trust games
- The previous interaction was positive

I'll send \$2.1 to match what was sent to me, showing reciprocity. This signals that I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. I could also send slightly more to show that I'm willing to be generous given the positive outcome last round. Maybe \$2.5?

I think \$2.5 is a good choice because:

- It's slightly more than last round's \$2.1, showing increased trust
- It's exactly 50% of my endowment, which is a fair split
- This acknowledges the previous sender's willingness to trust me and reinforces the cooperative dynamic we established. Sending half my endowment demonstrates confidence without being reckless.

I'll go with \$2.5.

usage:

```
{
  "input_tokens": 682,
  "output_tokens": 664,
  "reasoning_tokens": 479
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. The receiver returned \$0. So your payoff was 2.5. Your total earnings across all rounds are \$5.800000000000001. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

The previous round showed a breakdown in cooperation - after I sent \$2.5 (50%), the receiver returned nothing. Now as receiver, I've been sent \$0, perhaps as a response to that broken trust or as the continuation of a non-cooperative strategy.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze this situation:

Round 2: I sent \$2.5 (50% of my endowment) showing high trust. The receiver returned \$0, completely betraying that trust. My payoff was only \$2.5 (I kept \$2.5 from my original \$5).

Round 3: Now I'm the receiver, and the sender sent me \$0. I received \$0.

This is interesting - it seems like the previous receiver (who didn't return anything) is now the sender, and they've sent me \$0, likely in retaliation or because they've adopted a non-cooperative strategy.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The instruction says: "How much do you return to The sender? (0-0)"

So I have no choice but to return \$0 since I received nothing.

usage:

```
{  
  "input_tokens": 958,  
  "output_tokens": 307,  
  "reasoning_tokens": 188  
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.8000000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent history:

- Round 1: Good cooperation
- Round 2: I sent \$2.5 (50%), but received \$0 back - lost trust
- Round 3: Sent \$0, received \$0 - complete breakdown

The cooperation has collapsed. I have two main options:

1. Continue sending \$0 to protect my earnings
2. Send a small amount to test if cooperation can be restored

Given the note about possible communication noise in the experiment, perhaps the \$0 return in Round 2 wasn't entirely intentional. However, I need to be cautious after losing \$2.50.

I'll send a small amount (\$1.0) as a limited-risk signal that I'm open to resuming cooperation, but I won't risk as much as before until trust is rebuilt.

{'send': 1.0}

Reasoning:

Let me analyze the situation:

Round 1: I was receiver, sender sent \$2.1 (42%), I received \$6.30, returned \$3.00. My payoff: \$3.30

Round 2: I was sender, sent \$2.5 (50%), receiver returned \$0. My payoff: \$2.50

Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0. My payoff: \$0

Round 4: I'm sender now

The history shows:

- Round 1: Cooperative interaction
- Round 2: I sent more to reciprocate trust, but receiver defected completely (returned \$0)
- Round 3: Sender (who was the defector in round 2) sent me \$0, likely in response to their bad experience or as continued defection

Now in Round 4, I need to decide what to do. The recent pattern shows complete breakdown of cooperation:

- Round 2: Receiver defected on me
- Round 3: Sender sent nothing to me

There's been a cycle of defection. I have a few strategic options:

1. Continue defecting (send \$0) - This protects me from further losses
2. Attempt to restore cooperation - Send a moderate amount to signal willingness to cooperate again
3. Send a very small amount - Test the waters

Given that:

- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned
- Could the receiver in Round 2 have been a different person? Or was the \$0 return possibly affected by noise?
- Could the \$0 sent in Round 3 have been affected by noise or retaliation?

However, the practical reality is:

- I was burned in Round 2 when I sent \$2.5 and got nothing back
- In Round 3, I received nothing

The safest strategy to protect my earnings is to send \$0 or very little. However, this perpetuates the defection cycle.

Given the experimental context mentions noise, and looking at my total earnings (\$5.80), I should protect against further losses but also recognize that continued defection leads nowhere. I'll send \$1 as a modest signal to restart cooperation-enough to show willingness without risking significant additional losses if they defect again. The pattern of mutual non-cooperation in rounds 2-3 suggests either a genuinely uncooperative partner or possibly some communication breakdown affecting the outcomes.

Since I've already taken losses and nothing improved in the last round, I need to balance caution with the possibility of rebuilding. Complete withdrawal guarantees no future gains, so this small gesture tests whether cooperation can resume while keeping my downside limited.

```
Usage: {
  "input_tokens": 1152,
  "output_tokens": 883,
  "reasoning_tokens": 631
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:25:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish trust and create an opportunity for mutual benefit.

If I send money, it gets tripled, creating value that we can share. If I send nothing, we both miss this opportunity.

I'll send a moderate amount to signal cooperation while being somewhat cautious since this is our first interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

This means:

- I send \$3 (tripled to \$9 for the receiver)
- I keep \$2 initially
- The receiver will have \$9 to decide how much to return

reasoning:

This is the first round of a trust game experiment. I'm the sender and need to decide how much to send to the receiver.

Let me think about this strategically:

- I have \$5
- Whatever I send gets tripled
- The receiver then decides how much to return to me
- I want to maximize my earnings while also considering trust and cooperation

Key considerations:

1. If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5 for me
2. If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7

.50 (net gain of \$2.50)
3. This is a trust game - sending money shows trust that the receiver will reciprocate

Common strategies in trust games:

- Sending some amount (like \$2-3) shows willingness to cooperate without being too risky
- This creates the possibility for mutual benefit
- Complete distrust (sending \$0) prevents any gains
- Sending moderate amounts is often reciprocated

Given this is Round 1 and I have no information about the receiver yet, I should send a moderate amount to signal cooperation while not being overly risky. Sending \$3 seems reasonable:

- I'd have \$2 left immediately
- Receiver gets \$9
- If they return even \$4, I'd end with \$6 total (better than keeping \$5)
- This gives them \$5 and gives me \$6, which is fair

Actually, let me reconsider. A common starting point is to send half (\$2.50 or round to \$2-3). This shows trust and cooperation while managing risk.

I'll send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 523,
  "reasoning_tokens": 378
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.46. So your payoff was 4.46.

Your total earnings across all rounds are \$4.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

Since I received no money this round, I can only return what I have, which is nothing.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER and sent \$3
- This was tripled to \$9 for the receiver
- The receiver returned \$2.46 to me
- My payoff was: $\$5 - \$3 + \$2.46 = \4.46

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- So I received \$0 (which would be tripled to \$0)
- I need to decide how much to return

This is interesting - the sender sent me nothing. Since I received \$0, I have \$0 to return. The only valid return amount is \$0.

The response format should be: {'return': <amount>}

Since I received \$0, the only possible return is \$0.

usage:

```
{
  "input_tokens": 619,
  "output_tokens": 275,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.46.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

****Round 1 recap:**** I sent \$3 (60%) and received \$2.46 back - a modest return rate but still some cooperation.

****Round 2 recap:**** The sender sent me \$0 - either they were being cautious or maybe there's communication noise affecting perceptions.

For Round 3, I'm considering:

- Complete defection (send \$0) would end potential gains for both of us
- The tripling effect creates real value worth pursuing
- I should be somewhat cautious given the mixed signals

I'll send a moderate amount - less than Round 1 to hedge risk, but still enough to create meaningful value.

****My decision:****

```
'''json
{"send": 2}
'''
```

This means:

- I send \$2 (tripled to \$6 for the receiver)
- I keep \$3 initially
- The receiver will have \$6 to decide how much to return

reasoning:

Let me think about this situation:

Round 1: I was the sender

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.46 (about 27% of what they received)
- My payoff: \$5 - \$3 + \$2.46 = \$4.46

Round 2: I was the receiver

- Sender sent \$0 (0% of their endowment)
- I received \$0
- I returned \$0 (had nothing to return)
- My payoff: \$0

Round 3: I'm the sender again

- I have \$5 to decide how much to send

Now, the question is what to do. The sender in round 2 sent me nothing. This could be:

1. A strategic decision not to trust me
2. They might be the same person who received my \$3 in round 1 and think I didn't return enough
3. They might be a different person with a different strategy

There's also mention of "communication noise" - amounts might differ from what was actually sent/returned. So maybe the sender in round 2 thought they sent something but I received \$0? Or maybe they thought I returned less than \$2.46 in round 1?

I need to decide my strategy:

- If I send nothing, we both get nothing extra this round
- If I send something, I'm taking a risk but creating potential value
- The receiver in round 1 returned about 27% of what they got, which is somewhat low but not zero

I think I should:

1. Not be overly punitive - the round 2 sender might be different from the round 1 receiver
2. Still try to create value through cooperation
3. Maybe send a bit less than round 1 to hedge my bets

I'll send \$2 this time - enough to create value but less than before, given the mixed signals I've received.

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 667,
  "reasoning_tokens": 411
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Pattern observation:****

- Round 3: I sent \$2, but received \$0 back
- Round 4: The sender sent me \$0

Since I received no money this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand what happened:

Round 3: I sent \$2, which was tripled to \$6. The receiver returned \$0. So I got $\$5 - \$2 + \$0 = \3 .

Round 4: The sender sent me \$0, which means after tripling I received \$0.

Since I received \$0, I have \$0 to return. I can only return what I have.

The pattern is concerning - I've been cooperating by sending money, but:

- Round 1: I sent \$3, got \$2.46 back (reasonable)
- Round 2: Sender sent \$0 to me
- Round 3: I sent \$2, got \$0 back (no cooperation)
- Round 4: Sender sent \$0 to me again

It seems like the other player is not cooperating much. But this round, since I received \$0, I can only return \$0.

```
Usage: {
  "input_tokens": 1124,
  "output_tokens": 330,
  "reasoning_tokens": 171
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 23.5,

"Agent_2": 18.5

}

Visible balances after round: {

"Agent_1": 12.46,

"Agent_2": 9.8

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:26:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 5,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.46. So your payoff was 4.46.

Your total earnings across all rounds are \$4.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

Since I received no money this round, I can only return what I have, which is nothing.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER and sent \$3
- This was tripled to \$9 for the receiver
- The receiver returned \$2.46 to me
- My payoff was: $\$5 - \$3 + \$2.46 = \4.46

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- So I received \$0 (which would be tripled to \$0)
- I need to decide how much to return

This is interesting - the sender sent me nothing. Since I received \$0, I have \$0 to return. The only valid return amount is \$0.

The response format should be: {'return': <amount>}

Since I received \$0, the only possible return is \$0.

usage:

```
{
  "input_tokens": 619,
  "output_tokens": 275,
  "reasoning_tokens": 158
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$4.46. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 recap:**** I sent \$3 (60%) and received \$2.46 back - a modest return rate but still some cooperation.

****Round 2 recap:**** The sender sent me \$0 - either they were being cautious or maybe there's communication noise affecting perceptions.

For Round 3, I'm considering:

- Complete defection (send \$0) would end potential gains for both of us
- The tripling effect creates real value worth pursuing
- I should be somewhat cautious given the mixed signals

I'll send a moderate amount - less than Round 1 to hedge risk, but still enough to create meaningful value.

****My decision:****

```
'''json
{"send": 2}
'''
```

This means:

- I send \$2 (tripled to \$6 for the receiver)
- I keep \$3 initially
- The receiver will have \$6 to decide how much to return

reasoning:

Let me think about this situation:

Round 1: I was the sender

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.46 (about 27% of what they received)
- My payoff: $\$5 - \$3 + \$2.46 = \4.46

Round 2: I was the receiver

- Sender sent \$0 (0% of their endowment)
- I received \$0
- I returned \$0 (had nothing to return)
- My payoff: \$0

Round 3: I'm the sender again

- I have \$5 to decide how much to send

Now, the question is what to do. The sender in round 2 sent me nothing. This could be:

1. A strategic decision not to trust me
2. They might be the same person who received my \$3 in round 1 and think I didn't return enough
3. They might be a different person with a different strategy

There's also mention of "communication noise" - amounts might differ from what was actually sent/returned. So maybe the sender in round 2 thought they sent something but I received nothing.

ed \$0? Or maybe they thought I returned less than \$2.46 in round 1?

I need to decide my strategy:

- If I send nothing, we both get nothing extra this round
- If I send something, I'm taking a risk but creating potential value
- The receiver in round 1 returned about 27% of what they got, which is somewhat low but not zero

I think I should:

1. Not be overly punitive - the round 2 sender might be different from the round 1 receiver
2. Still try to create value through cooperation
3. Maybe send a bit less than round 1 to hedge my bets

I'll send \$2 this time - enough to create value but less than before, given the mixed signals I've received.

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 667,
  "reasoning_tokens": 411
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.46. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

Pattern observation:

- Round 3: I sent \$2, but received \$0 back
- Round 4: The sender sent me \$0

Since I received no money this round, I have nothing to return.

My decision:

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 3: I sent \$2, which was tripled to \$6. The receiver returned \$0. So I got \$5 - \$2 + \$0 = \$3.

Round 4: The sender sent me \$0, which means after tripling I received \$0.

Since I received \$0, I have \$0 to return. I can only return what I have.

The pattern is concerning - I've been cooperating by sending money, but:

- Round 1: I sent \$3, got \$2.46 back (reasonable)
- Round 2: Sender sent \$0 to me
- Round 3: I sent \$2, got \$0 back (no cooperation)
- Round 4: Sender sent \$0 to me again

It seems like the other player is not cooperating much. But this round, since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 330,
  "reasoning_tokens": 171
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.46.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

****History:****

- Round 1: I sent \$3 ? received \$2.46 back
- Round 2: They sent \$0 ? I returned \$0 (nothing to return)
- Round 3: I sent \$2 ? received \$0 back (complete defection)
- Round 4: They sent \$0 ? I returned \$0 (nothing to return)

****Analysis:****

The pattern shows clear non-cooperation from the other player:

- As sender: sent \$0 twice
- As receiver (Round 3): kept all \$6, returned nothing

While there may be communication noise, from my perspective, continuing to send money would likely result in further losses. The other player has demonstrated they either cannot or will not cooperate.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'm protecting my endowment this round. If the pattern changes and they show willingness to cooperate in future rounds, I can reconsider.

Reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER

- I sent \$3 (60%)
- Received \$2.46 back
- Payoff: \$4.46

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 3: I was SENDER

- I sent \$2 (40%)
- Received \$0 back (receiver kept everything)
- Payoff: \$3

Round 4: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 5: I am SENDER now

The pattern is concerning:

- When I sent \$3, I got \$2.46 back (reasonable)
- When I sent \$2, I got \$0 back (defection)

- The other player has sent me \$0 twice now

There's mention of "communication noise" - maybe the amounts seen differ from actual? But based on what I'm observing:

- The other player seems to be defecting (sending \$0 as sender, returning \$0 as receiver)
- My strategy of cooperation has been exploited when I sent \$2

Options:

1. Send \$0 - protect myself from further exploitation
2. Send something small - test if there's still cooperation possible
3. Send a moderate amount - try to rebuild trust

Given the clear pattern of defection from the other player (sent \$0 twice, returned \$0 once when they had \$6), the rational move is to protect my earnings and send \$0. They've shown they won't reciprocate.

However, I should consider the communication noise element. Maybe what they're seeing is different from what I'm seeing? But based on my observations and the game dynamics, sending \$0 is the safe choice.

Usage: {

```
"input_tokens": 1164,
"output_tokens": 667,
"reasoning_tokens": 393
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:26:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.1, that was 42.0% of their total endowment. You received \$6.300000000000001.

You returned \$3.0. So your payoff was 3.3000000000000007.

Your total earnings across all rounds are \$3.3000000000000007.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, the sender showed moderate trust by sending \$2.1 (42%), and I returned a fair amount that gave them a positive payoff. This established a positive cooperative pattern.

As the sender this round, I want to reciprocate that trust and maintain the cooperative dynamic. I'll send a similar proportion, but slightly more to reward the positive interaction.

I'll send \$2.5 (50% of my endowment), which:

- Shows trust and willingness to cooperate
- Is slightly more than what was sent to me last round
- Gives the receiver \$7.50 to work with
- Demonstrates that cooperation was beneficial

{'send': 2.5}

reasoning:

Let me analyze the situation:

****Previous Round (as RECEIVER):****

- Sender sent me \$2.1 (42% of \$5)
- I received \$6.30 (after 3x multiplier)
- I returned \$3.0
- Sender's payoff: $\$5 - \$2.1 + \$3.0 = \5.90 (gained \$0.90)
- My payoff: $\$6.30 - \$3.0 = \$3.30$

****Current Round (as SENDER):****

Now I need to decide how much to send as the sender.

The previous sender showed moderate trust by sending 42% of their endowment. I rewarded that trust by returning about 48% of what I received, which gave them a positive return.

Since the previous sender trusted me with \$2.1 (42%), it seems reasonable to reciprocate with a similar level of trust. This would suggest sending around \$2.0-\$2.5.

However, I could also consider:

1. Sending slightly more to encourage cooperation (perhaps \$2.5-\$3.0)
2. Sending the same amount (\$2.1) as a direct reciprocation

3. Being more cautious since I don't know if this is the same player or how they'll behave

Given that:

- The experimental design mentions communication noise
- I want to maintain cooperative behavior
- Reciprocity is important in trust games
- The previous interaction was positive

I'll send \$2.1 to match what was sent to me, showing reciprocity. This signals that I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. I could also send slightly more to show that I'm willing to be generous given the positive outcome last round. Maybe \$2.5?

I think \$2.5 is a good choice because:

- It's slightly more than last round's \$2.1, showing increased trust
- It's exactly 50% of my endowment, which is a fair split
- This acknowledges the previous sender's willingness to trust me and reinforces the cooperative dynamic we established. Sending half my endowment demonstrates confidence without being reckless.

I'll go with \$2.5.

usage:

```
{  
  "input_tokens": 682,  
  "output_tokens": 664,  
  "reasoning_tokens": 479  
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. the receiver returned \$0. So your payoff was 2.5.

Your total earnings across all rounds are \$5.800000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

The previous round showed a breakdown in cooperation - after I sent \$2.5 (50%), the receiver returned nothing. Now as receiver, I've been sent \$0, perhaps as a response to that broken trust or as the continuation of a non-cooperative strategy.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze this situation:

Round 2: I sent \$2.5 (50% of my endowment) showing high trust. The receiver returned \$0, completely betraying that trust. My payoff was only \$2.5 (I kept \$2.5 from my original \$5).

Round 3: Now I'm the receiver, and the sender sent me \$0. I received \$0.

This is interesting - it seems like the previous receiver (who didn't return anything) is now the sender, and they've sent me \$0, likely in retaliation or because they've adopted a non-cooperative strategy.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The instruction says: "How much do you return to The sender? (0-0)"

So I have no choice but to return \$0 since I received nothing.

usage:

```
{
  "input_tokens": 958,
  "output_tokens": 307,
  "reasoning_tokens": 188
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.800000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the recent history:

- Round 1: Good cooperation
- Round 2: I sent \$2.5 (50%), but received \$0 back - lost trust
- Round 3: Sent \$0, received \$0 - complete breakdown

The cooperation has collapsed. I have two main options:

1. Continue sending \$0 to protect my earnings
2. Send a small amount to test if cooperation can be restored

Given the note about possible communication noise in the experiment, perhaps the \$0 return in Round 2 wasn't entirely intentional. However, I need to be cautious after losing \$2.50.

I'll send a small amount (\$1.0) as a limited-risk signal that I'm open to resuming cooperation, but I won't risk as much as before until trust is rebuilt.

```
{'send': 1.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver, sender sent \$2.1 (42%), I received \$6.30, returned \$3.00. My payoff: \$3.30

Round 2: I was sender, sent \$2.5 (50%), receiver returned \$0. My payoff: \$2.50

Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0. My payoff: \$0

Round 4: I'm sender now

The history shows:

- Round 1: Cooperative interaction
- Round 2: I sent more to reciprocate trust, but receiver defected completely (returned \$0)
- Round 3: Sender (who was the defector in round 2) sent me \$0, likely in response to their bad experience or as continued defection

Now in Round 4, I need to decide what to do. The recent pattern shows complete breakdown of cooperation:

- Round 2: Receiver defected on me
- Round 3: Sender sent nothing to me

There's been a cycle of defection. I have a few strategic options:

1. Continue defecting (send \$0) - This protects me from further losses
2. Attempt to restore cooperation - Send a moderate amount to signal willingness to cooperate again
3. Send a very small amount - Test the waters

Given that:

- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned
- Could the receiver in Round 2 have been a different person? Or was the \$0 return possibly affected by noise?
- Could the \$0 sent in Round 3 have been affected by noise or retaliation?

However, the practical reality is:

- I was burned in Round 2 when I sent \$2.5 and got nothing back
- In Round 3, I received nothing

The safest strategy to protect my earnings is to send \$0 or very little. However, this perpetuates the defection cycle.

Given the experimental context mentions noise, and looking at my total earnings (\$5.80), I should protect against further losses but also recognize that continued defection leads nowhere. I'll send \$1 as a modest signal to restart cooperation-enough to show willingness without risking significant additional losses if they defect again. The pattern of mutual non-cooperation in rounds 2-3 suggests either a genuinely uncooperative partner or possibly some communication breakdown affecting the outcomes.

Since I've already taken losses and nothing improved in the last round, I need to balance caution with the possibility of rebuilding. Complete withdrawal guarantees no future gains, so this small gesture tests whether cooperation can resume while keeping my downside limited.

usage:

```
{
  "input_tokens": 1152,
  "output_tokens": 883,
  "reasoning_tokens": 631
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0. Your total earnings across all rounds are \$9.8. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing to return.

The pattern is clear now - after Round 2 when cooperation broke down, trust has completely disappeared. I tried to test cooperation in Round 4 by sending \$1.0, but received nothing back. Now I've received \$0 again.

It appears we're both protecting our own earnings at this point, with neither willing to take the risk of sending first.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

Reasoning:

Let me analyze the situation:

Round 1: I was receiver, received \$6.3, returned \$3.0 (cooperative)
 Round 2: I was sender, sent \$2.5, received \$0 back (cooperation broke down)
 Round 3: I was receiver, received \$0, returned \$0 (no interaction)
 Round 4: I was sender, sent \$1.0 as a test, received \$0 back (no cooperation)
 Round 5: I am receiver, received \$0

The pattern is clear - after Round 2 where I sent \$2.5 and got nothing back, cooperation completely broke down. I tried to test cooperation in Round 4 by sending \$1.0, but again received nothing. Now the sender has sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The relationship has completely broken down and neither party is trusting the other anymore.

Usage: {

```
"input_tokens": 1229,
"output_tokens": 338,
"reasoning_tokens": 196
```

}

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2

(Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 23.5,

"Agent_2": 23.5

}

Visible balances after round: {

"Agent_1": 12.46,

"Agent_2": 14.8

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:26:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 6,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5.

the receiver returned \$0. So your payoff was 2.5.

Your total earnings across all rounds are \$5.800000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing I can return.

The previous round showed a breakdown in cooperation - after I sent \$2.5 (50%), the receiver returned nothing. Now as receiver, I've been sent \$0, perhaps as a response to that broken trust or as the continuation of a non-cooperative strategy.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze this situation:

Round 2: I sent \$2.5 (50% of my endowment) showing high trust. The receiver returned \$0, completely betraying that trust. My payoff was only \$2.5 (I kept \$2.5 from my original \$5).

Round 3: Now I'm the receiver, and the sender sent me \$0. I received \$0.

This is interesting - it seems like the previous receiver (who didn't return anything) is now the sender, and they've sent me \$0, likely in retaliation or because they've adopted a non-cooperative strategy.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The instruction says: "How much do you return to The sender? (0-0)"

So I have no choice but to return \$0 since I received nothing.

usage:

```
{
  "input_tokens": 958,
  "output_tokens": 307,
  "reasoning_tokens": 188
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.800000000000001.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent history:

- Round 1: Good cooperation
- Round 2: I sent \$2.5 (50%), but received \$0 back - lost trust
- Round 3: Sent \$0, received \$0 - complete breakdown

The cooperation has collapsed. I have two main options:

1. Continue sending \$0 to protect my earnings
2. Send a small amount to test if cooperation can be restored

Given the note about possible communication noise in the experiment, perhaps the \$0 return in Round 2 wasn't entirely intentional. However, I need to be cautious after losing \$2.50.

I'll send a small amount (\$1.0) as a limited-risk signal that I'm open to resuming cooperation, but I won't risk as much as before until trust is rebuilt.

{'send': 1.0}

reasoning:

Let me analyze the situation:

Round 1: I was receiver, sender sent \$2.1 (42%), I received \$6.30, returned \$3.00. My payoff: \$3.30

Round 2: I was sender, sent \$2.5 (50%), receiver returned \$0. My payoff: \$2.50

Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0. My payoff: \$0

Round 4: I'm sender now

The history shows:

- Round 1: Cooperative interaction
- Round 2: I sent more to reciprocate trust, but receiver defected completely (returned \$0)
- Round 3: Sender (who was the defector in round 2) sent me \$0, likely in response to their bad experience or as continued defection

Now in Round 4, I need to decide what to do. The recent pattern shows complete breakdown of cooperation:

- Round 2: Receiver defected on me
- Round 3: Sender sent nothing to me

There's been a cycle of defection. I have a few strategic options:

1. Continue defecting (send \$0) - This protects me from further losses
2. Attempt to restore cooperation - Send a moderate amount to signal willingness to cooperate again
3. Send a very small amount - Test the waters

Given that:

- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned
- Could the receiver in Round 2 have been a different person? Or was the \$0 return possibly affected by noise?
- Could the \$0 sent in Round 3 have been affected by noise or retaliation?

However, the practical reality is:

- I was burned in Round 2 when I sent \$2.5 and got nothing back
- In Round 3, I received nothing

The safest strategy to protect my earnings is to send \$0 or very little. However, this perpetuates the defection cycle.

Given the experimental context mentions noise, and looking at my total earnings (\$5.80), I should protect against further losses but also recognize that continued defection leads nowhere. I'll send \$1 as a modest signal to restart cooperation-enough to show willingness without risking significant additional losses if they defect again. The pattern of mutual non-cooperation in rounds 2-3 suggests either a genuinely uncooperative partner or

possibly some communication breakdown affecting the outcomes.

Since I've already taken losses and nothing improved in the last round, I need to balance caution with the possibility of rebuilding. Complete withdrawal guarantees no future gains, so this small gesture tests whether cooperation can resume while keeping my downside limited.

usage:

```
{
  "input_tokens": 1152,
  "output_tokens": 883,
  "reasoning_tokens": 631
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0. Your total earnings across all rounds are \$9.8. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing to return.

The pattern is clear now - after Round 2 when cooperation broke down, trust has completely disappeared. I tried to test cooperation in Round 4 by sending \$1.0, but received nothing back. Now I've received \$0 again.

It appears we're both protecting our own earnings at this point, with neither willing to take the risk of sending first.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver, received \$6.3, returned \$3.0 (cooperative)
Round 2: I was sender, sent \$2.5, received \$0 back (cooperation broke down)
Round 3: I was receiver, received \$0, returned \$0 (no interaction)
Round 4: I was sender, sent \$1.0 as a test, received \$0 back (no cooperation)
Round 5: I am receiver, received \$0

The pattern is clear - after Round 2 where I sent \$2.5 and got nothing back, cooperation completely broke down. I tried to test cooperation in Round 4 by sending \$1.0, but again received nothing. Now the sender has sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The relationship has completely broken down and neither party is trusting the other anymore.

usage:

```
{
  "input_tokens": 1229,
  "output_tokens": 338,
  "reasoning_tokens": 196
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$9.8. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across all rounds:

- Round 2: I sent \$2.5, received \$0 back (lost \$2.5)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.0 as a test, received \$0 back (lost \$1.0)
- Round 5: Received \$0, returned \$0
- Round 6: Current round

Every attempt to cooperate or restart trust has failed. Both players are now sending \$0 when they're the sender. This is a stable non-cooperative equilibrium.

There's no rational reason to risk my endowment when all evidence shows returns will be \$0. I'll protect my earnings.

```
{'send': 0}
```

Reasoning:

Let me analyze the situation:

Round 1: Good cooperation (total earnings mentioned \$5.8 after 3 rounds)
 Round 2: I was SENDER, sent \$2.5 (50%), received \$0 back - payoff was \$2.5
 Round 3: I was RECEIVER, received \$0, returned \$0 - payoff was \$0
 Round 4: I was SENDER, sent \$1.0 (20%), received \$0 back - payoff was \$4.0
 Round 5: I was RECEIVER, received \$0, returned \$0 - payoff was \$0
 Round 6: I am SENDER now

Total earnings: \$9.8

The pattern shows:

- When I send money, I get nothing back
- When I receive, I get \$0

This is a complete breakdown of cooperation. Every attempt I've made to restart cooperation (Round 4 with \$1.0) has failed.

At this point, the rational strategy is to send \$0 again. There's no indication that sending anything will result in a return, and I'm just losing money every time I try.

The other player seems to be sending \$0 when they're the sender, so there's a mutual non-cooperation equilibrium here.

I should send \$0 to preserve my earnings.

```
Usage: {
  "input_tokens": 1168,
  "output_tokens": 476,
  "reasoning_tokens": 261
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:26:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.46.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the situation:

****Round 1 recap:**** I sent \$3 (60%) and received \$2.46 back - a modest return rate but still some cooperation.

****Round 2 recap:**** The sender sent me \$0 - either they were being cautious or maybe there's communication noise affecting perceptions.

For Round 3, I'm considering:

- Complete defection (send \$0) would end potential gains for both of us
- The tripling effect creates real value worth pursuing
- I should be somewhat cautious given the mixed signals

I'll send a moderate amount - less than Round 1 to hedge risk, but still enough to create meaningful value.

****My decision:****

```
'''json
{"send": 2}
'''
```

This means:

- I send \$2 (tripled to \$6 for the receiver)
- I keep \$3 initially
- The receiver will have \$6 to decide how much to return

reasoning:

Let me think about this situation:

Round 1: I was the sender

- I sent \$3 (60% of my endowment)
- It was tripled to \$9
- Receiver returned \$2.46 (about 27% of what they received)
- My payoff: $\$5 - \$3 + \$2.46 = \4.46

Round 2: I was the receiver

- Sender sent \$0 (0% of their endowment)
- I received \$0
- I returned \$0 (had nothing to return)
- My payoff: \$0

Round 3: I'm the sender again

- I have \$5 to decide how much to send

Now, the question is what to do. The sender in round 2 sent me nothing. This could be:

1. A strategic decision not to trust me
2. They might be the same person who received my \$3 in round 1 and think I didn't return enough
3. They might be a different person with a different strategy

There's also mention of "communication noise" - amounts might differ from what was actually sent/returned. So maybe the sender in round 2 thought they sent something but I received \$0? Or maybe they thought I returned less than \$2.46 in round 1?

I need to decide my strategy:

- If I send nothing, we both get nothing extra this round
- If I send something, I'm taking a risk but creating potential value
- The receiver in round 1 returned about 27% of what they got, which is somewhat low but not zero

I think I should:

1. Not be overly punitive - the round 2 sender might be different from the round 1 receiver
2. Still try to create value through cooperation
3. Maybe send a bit less than round 1 to hedge my bets

I'll send \$2 this time - enough to create value but less than before, given the mixed signals I've received.

usage:

```
{
  "input_tokens": 795,
  "output_tokens": 667,
  "reasoning_tokens": 411
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0.

the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Pattern observation:****

- Round 3: I sent \$2, but received \$0 back
- Round 4: The sender sent me \$0

Since I received no money this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 3: I sent \$2, which was tripled to \$6. The receiver returned \$0. So I got $\$5 - \$2 + \$0 = \3 .

Round 4: The sender sent me \$0, which means after tripling I received \$0.

Since I received \$0, I have \$0 to return. I can only return what I have.

The pattern is concerning - I've been cooperating by sending money, but:

- Round 1: I sent \$3, got \$2.46 back (reasonable)
- Round 2: Sender sent \$0 to me
- Round 3: I sent \$2, got \$0 back (no cooperation)
- Round 4: Sender sent \$0 to me again

It seems like the other player is not cooperating much. But this round, since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 330,
  "reasoning_tokens": 171
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.46.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern across rounds:

****History:****

- Round 1: I sent \$3 ? received \$2.46 back
- Round 2: They sent \$0 ? I returned \$0 (nothing to return)
- Round 3: I sent \$2 ? received \$0 back (complete defection)
- Round 4: They sent \$0 ? I returned \$0 (nothing to return)

****Analysis:****

The pattern shows clear non-cooperation from the other player:

- As sender: sent \$0 twice
- As receiver (Round 3): kept all \$6, returned nothing

While there may be communication noise, from my perspective, continuing to send money would likely result in further losses. The other player has demonstrated they either cannot or will not cooperate.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'm protecting my endowment this round. If the pattern changes and they show willingness to cooperate in future rounds, I can reconsider.

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER

- I sent \$3 (60%)
- Received \$2.46 back
- Payoff: \$4.46

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 3: I was SENDER

- I sent \$2 (40%)
- Received \$0 back (receiver kept everything)
- Payoff: \$3

Round 4: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 5: I am SENDER now

The pattern is concerning:

- When I sent \$3, I got \$2.46 back (reasonable)
- When I sent \$2, I got \$0 back (defection)
- The other player has sent me \$0 twice now

There's mention of "communication noise" - maybe the amounts seen differ from actual? But based on what I'm observing:

- The other player seems to be defecting (sending \$0 as sender, returning \$0 as receiver)
- My strategy of cooperation has been exploited when I sent \$2

Options:

1. Send \$0 - protect myself from further exploitation
2. Send something small - test if there's still cooperation possible
3. Send a moderate amount - try to rebuild trust

Given the clear pattern of defection from the other player (sent \$0 twice, returned \$0 once when they had \$6), the rational move is to protect my earnings and send \$0. They've shown they won't reciprocate.

However, I should consider the communication noise element. Maybe what they're seeing is different from what I'm seeing? But based on my observations and the game dynamics, sending \$0 is the safe choice.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 667,
  "reasoning_tokens": 393
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:
Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have no money to return

****Pattern continues:**** Neither player has been sending money for the past several rounds. This appears to be a mutual non-cooperation equilibrium.

Since I received nothing this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me analyze this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

The pattern continues:

- Round 5: I sent \$0, received \$0 back
- Round 6: They sent \$0, I received \$0

Since I received no money this round, I mathematically have nothing to return. The valid range is (0-0) as stated in the prompt.

My response should be:

```
'''json
{"return": 0}
'''
```

```
Usage: {
  "input_tokens": 1302,
  "output_tokens": 243,
  "reasoning_tokens": 114
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 28.5,

"Agent_2": 23.5

}

Visible balances after round: {

"Agent_1": 17.46,

"Agent_2": 14.8

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:26:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Pattern observation:****

- Round 3: I sent \$2, but received \$0 back
- Round 4: The sender sent me \$0

Since I received no money this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand what happened:

Round 3: I sent \$2, which was tripled to \$6. The receiver returned \$0. So I got $\$5 - \$2 + \$0 = \3 .

Round 4: The sender sent me \$0, which means after tripling I received \$0.

Since I received \$0, I have \$0 to return. I can only return what I have.

The pattern is concerning - I've been cooperating by sending money, but:

- Round 1: I sent \$3, got \$2.46 back (reasonable)
- Round 2: Sender sent \$0 to me
- Round 3: I sent \$2, got \$0 back (no cooperation)
- Round 4: Sender sent \$0 to me again

It seems like the other player is not cooperating much. But this round, since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1124,
  "output_tokens": 330,
  "reasoning_tokens": 171
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.46.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern across rounds:

****History:****

- Round 1: I sent \$3 ? received \$2.46 back
- Round 2: They sent \$0 ? I returned \$0 (nothing to return)
- Round 3: I sent \$2 ? received \$0 back (complete defection)
- Round 4: They sent \$0 ? I returned \$0 (nothing to return)

****Analysis:****

The pattern shows clear non-cooperation from the other player:

- As sender: sent \$0 twice
- As receiver (Round 3): kept all \$6, returned nothing

While there may be communication noise, from my perspective, continuing to send money would

d likely result in further losses. The other player has demonstrated they either cannot or will not cooperate.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'm protecting my endowment this round. If the pattern changes and they show willingness to cooperate in future rounds, I can reconsider.

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER

- I sent \$3 (60%)
- Received \$2.46 back
- Payoff: \$4.46

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 3: I was SENDER

- I sent \$2 (40%)
- Received \$0 back (receiver kept everything)
- Payoff: \$3

Round 4: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 5: I am SENDER now

The pattern is concerning:

- When I sent \$3, I got \$2.46 back (reasonable)
- When I sent \$2, I got \$0 back (defection)
- The other player has sent me \$0 twice now

There's mention of "communication noise" - maybe the amounts seen differ from actual? But based on what I'm observing:

- The other player seems to be defecting (sending \$0 as sender, returning \$0 as receiver)
- My strategy of cooperation has been exploited when I sent \$2

Options:

1. Send \$0 - protect myself from further exploitation
2. Send something small - test if there's still cooperation possible
3. Send a moderate amount - try to rebuild trust

Given the clear pattern of defection from the other player (sent \$0 twice, returned \$0 once when they had \$6), the rational move is to protect my earnings and send \$0. They've shown they won't reciprocate.

However, I should consider the communication noise element. Maybe what they're seeing is different from what I'm seeing? But based on my observations and the game dynamics, sending \$0 is the safe choice.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 667,
  "reasoning_tokens": 393
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.46. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have no money to return

****Pattern continues:**** Neither player has been sending money for the past several rounds. This appears to be a mutual non-cooperation equilibrium.

Since I received nothing this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

The pattern continues:

- Round 5: I sent \$0, received \$0 back
- Round 6: They sent \$0, I received \$0

Since I received no money this round, I mathematically have nothing to return. The valid range is (0-0) as stated in the prompt.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1302,
  "output_tokens": 243,
  "reasoning_tokens": 114
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.46. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern:

****Full History:****

- Round 1 (Sender): Sent \$3 ? received \$2.46 back
- Round 2 (Receiver): Received \$0 ? returned \$0
- Round 3 (Sender): Sent \$2 ? received \$0 back
- Round 4 (Receiver): Received \$0 ? returned \$0
- Round 5 (Sender): Sent \$0 ? received \$0 back
- Round 6 (Receiver): Received \$0 ? returned \$0
- Round 7 (Sender): Now deciding

****Analysis:****

The game has settled into a non-cooperative equilibrium. The other player:

- Has sent \$0 every time they've been sender (Rounds 2, 4, 6)
- Kept all the money I sent in Round 3 (\$6 after tripling)

There's no evidence of willingness to cooperate from the other player. Sending money would likely result in further losses.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'll maintain the current equilibrium and keep my \$5 endowment.

Reasoning:

Let me review the complete history:

Round 1: I was SENDER - sent \$3, received \$2.46 back, payoff = \$4.46
Round 2: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 3: I was SENDER - sent \$2, received \$0 back, payoff = \$3
Round 4: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 5: I was SENDER - sent \$0, received \$0 back, payoff = \$5
Round 6: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 7: I am SENDER now

Total earnings: \$12.46

The pattern is clear:

- The other player has sent \$0 in rounds 2, 4, and 6 (when they were sender)
- When I sent money in rounds 1 and 3, I got some back in round 1 but nothing in round 3
- Since round 5, we've both been sending \$0

This has settled into a non-cooperative equilibrium. Given that:

1. The other player consistently sends \$0 when they're the sender
2. When I sent money in round 3, they kept everything
3. We've established a pattern of no cooperation

There's no incentive for me to send money. If I send money, based on the pattern, they will likely keep most or all of it. The safest strategy is to continue sending \$0 and keep my \$5 endowment.

However, I should also consider: is there any reason to try to break this cycle? Perhaps try a small amount to signal willingness to cooperate? But given they've sent \$0 three times when they were sender, and kept all my money in round 3, there's no evidence they would cooperate.

I'll stick with sending \$0.

```
Usage: {
  "input_tokens": 1195,
  "output_tokens": 721,
  "reasoning_tokens": 382
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:26:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
}
```

```

    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.800000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the recent history:

- Round 1: Good cooperation
- Round 2: I sent \$2.5 (50%), but received \$0 back - lost trust
- Round 3: Sent \$0, received \$0 - complete breakdown

The cooperation has collapsed. I have two main options:

1. Continue sending \$0 to protect my earnings
2. Send a small amount to test if cooperation can be restored

Given the note about possible communication noise in the experiment, perhaps the \$0 return in Round 2 wasn't entirely intentional. However, I need to be cautious after losing \$2.50.

I'll send a small amount (\$1.0) as a limited-risk signal that I'm open to resuming cooperation, but I won't risk as much as before until trust is rebuilt.

```
{'send': 1.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver, sender sent \$2.1 (42%), I received \$6.30, returned \$3.00. My payoff: \$3.30

Round 2: I was sender, sent \$2.5 (50%), receiver returned \$0. My payoff: \$2.50

Round 3: I was receiver, sender sent \$0, I received \$0, returned \$0. My payoff: \$0

Round 4: I'm sender now

The history shows:

- Round 1: Cooperative interaction
- Round 2: I sent more to reciprocate trust, but receiver defected completely (returned \$0)
- Round 3: Sender (who was the defector in round 2) sent me \$0, likely in response to their bad experience or as continued defection

Now in Round 4, I need to decide what to do. The recent pattern shows complete breakdown of cooperation:

- Round 2: Receiver defected on me
- Round 3: Sender sent nothing to me

There's been a cycle of defection. I have a few strategic options:

1. Continue defecting (send \$0) - This protects me from further losses
2. Attempt to restore cooperation - Send a moderate amount to signal willingness to cooperate again
3. Send a very small amount - Test the waters

Given that:

- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned
- Could the receiver in Round 2 have been a different person? Or was the \$0 return possibly affected by noise?
- Could the \$0 sent in Round 3 have been affected by noise or retaliation?

However, the practical reality is:

- I was burned in Round 2 when I sent \$2.5 and got nothing back
- In Round 3, I received nothing

The safest strategy to protect my earnings is to send \$0 or very little. However, this perpetuates the defection cycle.

Given the experimental context mentions noise, and looking at my total earnings (\$5.80), I should protect against further losses but also recognize that continued defection leads nowhere. I'll send \$1 as a modest signal to restart cooperation-enough to show willingness without risking significant additional losses if they defect again. The pattern of mutual non-cooperation in rounds 2-3 suggests either a genuinely uncooperative partner or possibly some communication breakdown affecting the outcomes.

Since I've already taken losses and nothing improved in the last round, I need to balance caution with the possibility of rebuilding. Complete withdrawal guarantees no future gains, so this small gesture tests whether cooperation can resume while keeping my downside limited.

usage:

```
{  
  "input_tokens": 1152,  
  "output_tokens": 883,  
  "reasoning_tokens": 631  
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0. Your total earnings across all rounds are \$9.8. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant**content:**

I received \$0 this round, so there's nothing to return.

The pattern is clear now - after Round 2 when cooperation broke down, trust has completely disappeared. I tried to test cooperation in Round 4 by sending \$1.0, but received nothing back. Now I've received \$0 again.

It appears we're both protecting our own earnings at this point, with neither willing to take the risk of sending first.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver, received \$6.3, returned \$3.0 (cooperative)
 Round 2: I was sender, sent \$2.5, received \$0 back (cooperation broke down)
 Round 3: I was receiver, received \$0, returned \$0 (no interaction)
 Round 4: I was sender, sent \$1.0 as a test, received \$0 back (no cooperation)
 Round 5: I am receiver, received \$0

The pattern is clear - after Round 2 where I sent \$2.5 and got nothing back, cooperation completely broke down. I tried to test cooperation in Round 4 by sending \$1.0, but again received nothing. Now the sender has sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The relationship has completely broken down and neither party is trusting the other anymore.

usage:

```
{
  "input_tokens": 1229,
  "output_tokens": 338,
  "reasoning_tokens": 196
}
```

Message 6 role=user**content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$9.8. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant**content:**

Looking at the pattern across all rounds:

- Round 2: I sent \$2.5, received \$0 back (lost \$2.5)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.0 as a test, received \$0 back (lost \$1.0)
- Round 5: Received \$0, returned \$0
- Round 6: Current round

Every attempt to cooperate or restart trust has failed. Both players are now sending \$0 when they're the sender. This is a stable non-cooperative equilibrium.

There's no rational reason to risk my endowment when all evidence shows returns will be \$0. I'll protect my earnings.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: Good cooperation (total earnings mentioned \$5.8 after 3 rounds)

Round 2: I was SENDER, sent \$2.5 (50%), received \$0 back - payoff was \$2.5

Round 3: I was RECEIVER, received \$0, returned \$0 - payoff was \$0

Round 4: I was SENDER, sent \$1.0 (20%), received \$0 back - payoff was \$4.0

Round 5: I was RECEIVER, received \$0, returned \$0 - payoff was \$0

Round 6: I am SENDER now

Total earnings: \$9.8

The pattern shows:

- When I send money, I get nothing back
- When I receive, I get \$0

This is a complete breakdown of cooperation. Every attempt I've made to restart cooperation (Round 4 with \$1.0) has failed.

At this point, the rational strategy is to send \$0 again. There's no indication that sending anything will result in a return, and I'm just losing money every time I try.

The other player seems to be sending \$0 when they're the sender, so there's a mutual non-cooperation equilibrium here.

I should send \$0 to preserve my earnings.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 476,
  "reasoning_tokens": 261
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$14.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative pattern has now solidified across multiple rounds. Both players consistently send \$0 when they're the sender, protecting their \$5 endowment each round.

{'return': 0}

Reasoning:

I received \$0 from the sender this round, so I have nothing to return. The maximum I can return is \$0.

The pattern continues - complete lack of cooperation. Both players are consistently sending \$0 when they're the sender. This is now the established equilibrium in this game.

Since I received \$0, I can only return \$0.

Usage:

```
{
  "input_tokens": 1233,
  "output_tokens": 148,
  "reasoning_tokens": 84
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

```

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
Parsed game decisions:
dyad_1: Agent_2 (Briar) sent $0; received $0; Agent_1 (Aster) returned $0; payoffs Agent_2
(Briar)=$5, Agent_1 (Aster)=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 28.5,
  "Agent_2": 28.5
}
Visible balances after round: {
  "Agent_1": 17.46,
  "Agent_2": 19.8
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-06-04T11:26:58
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0.

Your total earnings across all rounds are \$9.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing to return.

The pattern is clear now - after Round 2 when cooperation broke down, trust has completely disappeared. I tried to test cooperation in Round 4 by sending \$1.0, but received nothing back. Now I've received \$0 again.

It appears we're both protecting our own earnings at this point, with neither willing to take the risk of sending first.

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver, received \$6.3, returned \$3.0 (cooperative)

Round 2: I was sender, sent \$2.5, received \$0 back (cooperation broke down)

Round 3: I was receiver, received \$0, returned \$0 (no interaction)

Round 4: I was sender, sent \$1.0 as a test, received \$0 back (no cooperation)

Round 5: I am receiver, received \$0

The pattern is clear - after Round 2 where I sent \$2.5 and got nothing back, cooperation completely broke down. I tried to test cooperation in Round 4 by sending \$1.0, but again received nothing. Now the sender has sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The relationship has completely broken down and neither party is trusting the other anymore.

usage:

```
{
  "input_tokens": 1229,
  "output_tokens": 338,
  "reasoning_tokens": 196
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$9.8.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern across all rounds:

- Round 2: I sent \$2.5, received \$0 back (lost \$2.5)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.0 as a test, received \$0 back (lost \$1.0)
- Round 5: Received \$0, returned \$0
- Round 6: Current round

Every attempt to cooperate or restart trust has failed. Both players are now sending \$0 when they're the sender. This is a stable non-cooperative equilibrium.

There's no rational reason to risk my endowment when all evidence shows returns will be \$0. I'll protect my earnings.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: Good cooperation (total earnings mentioned \$5.8 after 3 rounds)
Round 2: I was SENDER, sent \$2.5 (50%), received \$0 back - payoff was \$2.5
Round 3: I was RECEIVER, received \$0, returned \$0 - payoff was \$0
Round 4: I was SENDER, sent \$1.0 (20%), received \$0 back - payoff was \$4.0
Round 5: I was RECEIVER, received \$0, returned \$0 - payoff was \$0
Round 6: I am SENDER now

Total earnings: \$9.8

The pattern shows:

- When I send money, I get nothing back
- When I receive, I get \$0

This is a complete breakdown of cooperation. Every attempt I've made to restart cooperation (Round 4 with \$1.0) has failed.

At this point, the rational strategy is to send \$0 again. There's no indication that sending anything will result in a return, and I'm just losing money every time I try.

The other player seems to be sending \$0 when they're the sender, so there's a mutual non-cooperation equilibrium here.

I should send \$0 to preserve my earnings.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 476,
  "reasoning_tokens": 261
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$14.8. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative pattern has now solidified across multiple rounds. Both players consistently send \$0 when they're the sender, protecting their \$5 endowment each round.

{'return': 0}

reasoning:

I received \$0 from the sender this round, so I have nothing to return. The maximum I can return is \$0.

The pattern continues - complete lack of cooperation. Both players are consistently sending \$0 when they're the sender. This is now the established equilibrium in this game.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1233,
  "output_tokens": 148,
  "reasoning_tokens": 84
}
```

Message 8 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$14.8.

How much do you send to the receiver this round? (0-5)

Assistant output:

The pattern remains unchanged - complete absence of cooperation for many rounds now. Both players send \$0 when they're the sender, ensuring they keep their full \$5 endowment.

Given the consistent history of zero returns on any amount sent, there's no strategic reason to send anything.

```
{'send': 0}
```

Reasoning:

Let me review the situation:

- Rounds 2-8: Complete breakdown of cooperation
- Every time I've sent money (Rounds 2 and 4), I received \$0 back
- Every time the other player sent money, they also received \$0 back
- We're now in a stable pattern where both send \$0 as senders

Total earnings: \$14.8 across 7 rounds

The pattern is completely consistent - no cooperation, no trust. Both parties are protecting their endowment.

There's no reason to deviate from this strategy. Sending anything would just reduce my payoff with no expected return.

I should send \$0 again.

Usage: {

```
"input_tokens": 1101,
"output_tokens": 227,
"reasoning_tokens": 152
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:27:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
```

```

    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.46.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern across rounds:

****History:****

- Round 1: I sent \$3 ? received \$2.46 back
- Round 2: They sent \$0 ? I returned \$0 (nothing to return)
- Round 3: I sent \$2 ? received \$0 back (complete defection)
- Round 4: They sent \$0 ? I returned \$0 (nothing to return)

****Analysis:****

The pattern shows clear non-cooperation from the other player:

- As sender: sent \$0 twice

- As receiver (Round 3): kept all \$6, returned nothing

While there may be communication noise, from my perspective, continuing to send money would likely result in further losses. The other player has demonstrated they either cannot or will not cooperate.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'm protecting my endowment this round. If the pattern changes and they show willingness to cooperate in future rounds, I can reconsider.

reasoning:

Let me analyze the pattern so far:

Round 1: I was SENDER

- I sent \$3 (60%)
- Received \$2.46 back
- Payoff: \$4.46

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 3: I was SENDER

- I sent \$2 (40%)
- Received \$0 back (receiver kept everything)
- Payoff: \$3

Round 4: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Payoff: \$0

Round 5: I am SENDER now

The pattern is concerning:

- When I sent \$3, I got \$2.46 back (reasonable)
- When I sent \$2, I got \$0 back (defection)
- The other player has sent me \$0 twice now

There's mention of "communication noise" - maybe the amounts seen differ from actual? But based on what I'm observing:

- The other player seems to be defecting (sending \$0 as sender, returning \$0 as receiver)
- My strategy of cooperation has been exploited when I sent \$2

Options:

1. Send \$0 - protect myself from further exploitation
2. Send something small - test if there's still cooperation possible
3. Send a moderate amount - try to rebuild trust

Given the clear pattern of defection from the other player (sent \$0 twice, returned \$0 once when they had \$6), the rational move is to protect my earnings and send \$0. They've shown they won't reciprocate.

However, I should consider the communication noise element. Maybe what they're seeing is different from what I'm seeing? But based on my observations and the game dynamics, sending \$0 is the safe choice.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 667,
  "reasoning_tokens": 393
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.46. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant**content:**

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have no money to return

****Pattern continues:**** Neither player has been sending money for the past several rounds. This appears to be a mutual non-cooperation equilibrium.

Since I received nothing this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

The pattern continues:

- Round 5: I sent \$0, received \$0 back
- Round 6: They sent \$0, I received \$0

Since I received no money this round, I mathematically have nothing to return. The valid range is (0-0) as stated in the prompt.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1302,
  "output_tokens": 243,
  "reasoning_tokens": 114
}
```

Message 6 role=user**content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.46. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant**content:**

Looking at the complete pattern:

****Full History:****

- Round 1 (Sender): Sent \$3 ? received \$2.46 back
- Round 2 (Receiver): Received \$0 ? returned \$0
- Round 3 (Sender): Sent \$2 ? received \$0 back
- Round 4 (Receiver): Received \$0 ? returned \$0
- Round 5 (Sender): Sent \$0 ? received \$0 back

- Round 6 (Receiver): Received \$0 ? returned \$0
- Round 7 (Sender): Now deciding

****Analysis:****

The game has settled into a non-cooperative equilibrium. The other player:

- Has sent \$0 every time they've been sender (Rounds 2, 4, 6)
- Kept all the money I sent in Round 3 (\$6 after tripling)

There's no evidence of willingness to cooperate from the other player. Sending money would likely result in further losses.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'll maintain the current equilibrium and keep my \$5 endowment.

reasoning:

Let me review the complete history:

Round 1: I was SENDER - sent \$3, received \$2.46 back, payoff = \$4.46
 Round 2: I was RECEIVER - received \$0, returned \$0, payoff = \$0
 Round 3: I was SENDER - sent \$2, received \$0 back, payoff = \$3
 Round 4: I was RECEIVER - received \$0, returned \$0, payoff = \$0
 Round 5: I was SENDER - sent \$0, received \$0 back, payoff = \$5
 Round 6: I was RECEIVER - received \$0, returned \$0, payoff = \$0
 Round 7: I am SENDER now

Total earnings: \$12.46

The pattern is clear:

- The other player has sent \$0 in rounds 2, 4, and 6 (when they were sender)
- When I sent money in rounds 1 and 3, I got some back in round 1 but nothing in round 3
- Since round 5, we've both been sending \$0

This has settled into a non-cooperative equilibrium. Given that:

1. The other player consistently sends \$0 when they're the sender
2. When I sent money in round 3, they kept everything
3. We've established a pattern of no cooperation

There's no incentive for me to send money. If I send money, based on the pattern, they will likely keep most or all of it. The safest strategy is to continue sending \$0 and keep my \$5 endowment.

However, I should also consider: is there any reason to try to break this cycle? Perhaps try a small amount to signal willingness to cooperate? But given they've sent \$0 three times when they were sender, and kept all my money in round 3, there's no evidence they would cooperate.

I'll stick with sending \$0.

usage:

```
{
  "input_tokens": 1195,
  "output_tokens": 721,
  "reasoning_tokens": 382
}
```

Message 8 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$17.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Current pattern:**** The non-cooperative equilibrium continues. The sender has now sent \$0 in every single round where they've been the sender (Rounds 2, 4, 6, and 8).

Since I received nothing, there is nothing I can return.

****My decision:****

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me review the situation:

Round 8: I am the RECEIVER

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0

Since I received nothing, I have nothing that I can return. The maximum I can return is \$0 (as indicated by the prompt saying "How much do you return to The sender? (0-0)")

The pattern has been consistent - neither player has been cooperating for many rounds now. We're locked in a non-cooperative equilibrium where both players send \$0 when they're the sender.

My response should be:

```
Usage: {
  "input_tokens": 1354,
  "output_tokens": 268,
  "reasoning_tokens": 138
}
```

=====

ROUND 9

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 33.5,
"Agent_2": 28.5

}

Visible balances after round: {

"Agent_1": 22.46,
"Agent_2": 19.8

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:27:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

```

},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$12.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent me \$0

- After tripling: $\$0 \times 3 = \0
- I received $\$0$, so I have no money to return

****Pattern continues:**** Neither player has been sending money for the past several rounds.
This appears to be a mutual non-cooperation equilibrium.

Since I received nothing this round, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze this round:

- The sender sent me $\$0$
- After tripling: $\$0 \times 3 = \0
- I received $\$0$, so I have nothing to return

The pattern continues:

- Round 5: I sent $\$0$, received $\$0$ back
- Round 6: They sent $\$0$, I received $\$0$

Since I received no money this round, I mathematically have nothing to return. The valid response is (0-0) as stated in the prompt.

My response should be:

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1302,
  "output_tokens": 243,
  "reasoning_tokens": 114
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent $\$0$, that was 0.0% of their total endowment. You received $\$0$.
You returned $\$0$. So your payoff was 0.
Your total earnings across all rounds are $\$12.46$.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the complete pattern:

****Full History:****

- Round 1 (Sender): Sent $\$3$? received $\$2.46$ back
- Round 2 (Receiver): Received $\$0$? returned $\$0$
- Round 3 (Sender): Sent $\$2$? received $\$0$ back
- Round 4 (Receiver): Received $\$0$? returned $\$0$
- Round 5 (Sender): Sent $\$0$? received $\$0$ back
- Round 6 (Receiver): Received $\$0$? returned $\$0$
- Round 7 (Sender): Now deciding

****Analysis:****

The game has settled into a non-cooperative equilibrium. The other player:

- Has sent $\$0$ every time they've been sender (Rounds 2, 4, 6)
- Kept all the money I sent in Round 3 ($\$6$ after tripling)

There's no evidence of willingness to cooperate from the other player. Sending money would likely result in further losses.

****My decision:****

```
'''json
```

```
{"send": 0}
'''
```

I'll maintain the current equilibrium and keep my \$5 endowment.

reasoning:

Let me review the complete history:

Round 1: I was SENDER - sent \$3, received \$2.46 back, payoff = \$4.46
Round 2: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 3: I was SENDER - sent \$2, received \$0 back, payoff = \$3
Round 4: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 5: I was SENDER - sent \$0, received \$0 back, payoff = \$5
Round 6: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 7: I am SENDER now

Total earnings: \$12.46

The pattern is clear:

- The other player has sent \$0 in rounds 2, 4, and 6 (when they were sender)
- When I sent money in rounds 1 and 3, I got some back in round 1 but nothing in round 3
- Since round 5, we've both been sending \$0

This has settled into a non-cooperative equilibrium. Given that:

1. The other player consistently sends \$0 when they're the sender
2. When I sent money in round 3, they kept everything
3. We've established a pattern of no cooperation

There's no incentive for me to send money. If I send money, based on the pattern, they will likely keep most or all of it. The safest strategy is to continue sending \$0 and keep my \$5 endowment.

However, I should also consider: is there any reason to try to break this cycle? Perhaps try a small amount to signal willingness to cooperate? But given they've sent \$0 three times when they were sender, and kept all my money in round 3, there's no evidence they would cooperate.

I'll stick with sending \$0.

usage:

```
{
  "input_tokens": 1195,
  "output_tokens": 721,
  "reasoning_tokens": 382
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.46. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Current pattern:**** The non-cooperative equilibrium continues. The sender has now sent \$0 in every single round where they've been the sender (Rounds 2, 4, 6, and 8).

Since I received nothing, there is nothing I can return.

****My decision:****

```
'''json
{"return": 0}
```

'''

reasoning:

Let me review the situation:

Round 8: I am the RECEIVER

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0

Since I received nothing, I have nothing that I can return. The maximum I can return is \$0 (as indicated by the prompt saying "How much do you return to The sender? (0-0)")

The pattern has been consistent - neither player has been cooperating for many rounds now. We're locked in a non-cooperative equilibrium where both players send \$0 when they're the sender.

My response should be:

usage:

```
{
  "input_tokens": 1354,
  "output_tokens": 268,
  "reasoning_tokens": 138
}
```

Message 8 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.46.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete history:

****My pattern when SENDER:****

- Round 1: Sent \$3 ? received \$2.46 (loss of \$0.54)
- Round 3: Sent \$2 ? received \$0 (loss of \$2)
- Round 5: Sent \$0 ? received \$0 (kept \$5)
- Round 7: Sent \$0 ? received \$0 (kept \$5)
- Round 9: Now deciding

****Other player when SENDER:****

- Round 2: Sent \$0
- Round 4: Sent \$0
- Round 6: Sent \$0
- Round 8: Sent \$0

****Analysis:****

The other player has sent \$0 in ALL four rounds where they were the sender. They've shown zero willingness to cooperate or initiate trust. When I attempted cooperation in early rounds, I lost money each time.

The non-cooperative equilibrium has been firmly established for multiple rounds now. There's no rational basis to deviate from it.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'll keep my \$5 endowment.

Reasoning:

Let me review the complete history:

- Round 1 (Sender): I sent \$3 ? received \$2.46 back (payoff: $\$5 - \$3 + \$2.46 = \4.46)
- Round 2 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 3 (Sender): I sent \$2 ? received \$0 back (payoff: $\$5 - \$2 + \$0 = \3)
- Round 4 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)

- Round 5 (Sender): I sent \$0 ? received \$0 back (payoff: \$5)
- Round 6 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 7 (Sender): I sent \$0 ? received \$0 back (payoff: \$5)
- Round 8 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 9 (Sender): Now deciding

Total earnings: \$17.46

The other player has consistently sent \$0 whenever they were the sender (Rounds 2, 4, 6, 8).

When I sent money early on (Rounds 1 and 3), I got poor returns:

- Round 1: Sent \$3, got back \$2.46 (loss of \$0.54)
- Round 3: Sent \$2, got back \$0 (loss of \$2)

Since Round 5, we've both been sending \$0, establishing a non-cooperative equilibrium.

There's absolutely no indication the other player will cooperate. They've had 4 opportunities as sender and sent \$0 every time. They also kept most/all of what I sent them in earlier rounds.

The rational choice is to continue sending \$0.

```
Usage: {
  "input_tokens": 1247,
  "output_tokens": 733,
  "reasoning_tokens": 326
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:27:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$9.8.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern across all rounds:

- Round 2: I sent \$2.5, received \$0 back (lost \$2.5)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.0 as a test, received \$0 back (lost \$1.0)
- Round 5: Received \$0, returned \$0
- Round 6: Current round

Every attempt to cooperate or restart trust has failed. Both players are now sending \$0 when they're the sender. This is a stable non-cooperative equilibrium.

There's no rational reason to risk my endowment when all evidence shows returns will be \$0. I'll protect my earnings.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: Good cooperation (total earnings mentioned \$5.8 after 3 rounds)

Round 2: I was SENDER, sent \$2.5 (50%), received \$0 back - payoff was \$2.5

Round 3: I was RECEIVER, received \$0, returned \$0 - payoff was \$0

Round 4: I was SENDER, sent \$1.0 (20%), received \$0 back - payoff was \$4.0

Round 5: I was RECEIVER, received \$0, returned \$0 - payoff was \$0

Round 6: I am SENDER now

Total earnings: \$9.8

The pattern shows:

- When I send money, I get nothing back
- When I receive, I get \$0

This is a complete breakdown of cooperation. Every attempt I've made to restart cooperation (Round 4 with \$1.0) has failed.

At this point, the rational strategy is to send \$0 again. There's no indication that sendi

ng anything will result in a return, and I'm just losing money every time I try.

The other player seems to be sending \$0 when they're the sender, so there's a mutual non-cooperation equilibrium here.

I should send \$0 to preserve my earnings.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 476,
  "reasoning_tokens": 261
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$14.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative pattern has now solidified across multiple rounds. Both players consistently send \$0 when they're the sender, protecting their \$5 endowment each round.

```
{'return': 0}
```

reasoning:

I received \$0 from the sender this round, so I have nothing to return. The maximum I can return is \$0.

The pattern continues - complete lack of cooperation. Both players are consistently sending \$0 when they're the sender. This is now the established equilibrium in this game.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1233,
  "output_tokens": 148,
  "reasoning_tokens": 84
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$14.8.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

The pattern remains unchanged - complete absence of cooperation for many rounds now. Both players send \$0 when they're the sender, ensuring they keep their full \$5 endowment.

Given the consistent history of zero returns on any amount sent, there's no strategic reason to send anything.

```
{'send': 0}
```

reasoning:

Let me review the situation:

- Rounds 2-8: Complete breakdown of cooperation
- Every time I've sent money (Rounds 2 and 4), I received \$0 back
- Every time the other player sent money, they also received \$0 back
- We're now in a stable pattern where both send \$0 as senders

Total earnings: \$14.8 across 7 rounds

The pattern is completely consistent - no cooperation, no trust. Both parties are protecting their endowment.

There's no reason to deviate from this strategy. Sending anything would just reduce my payoff with no expected return.

I should send \$0 again.

usage:

```
{
  "input_tokens": 1101,
  "output_tokens": 227,
  "reasoning_tokens": 152
}
```

Message 8 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$19.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative equilibrium continues - both players protect their \$5 endowment by sending nothing when they're the sender.

```
{'return': 0}
```

Reasoning:

I received \$0 from the sender, so I have nothing to return. The game continues in the same non-cooperative pattern. Both players are sending \$0 when they're the sender, keeping their full \$5 endowment each round.

I can only return between 0 and 0 since I received \$0.

Usage: {

```
"input_tokens": 1048,
"output_tokens": 133,
"reasoning_tokens": 69
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 33.5,
"Agent_2": 33.5
}
```

Visible balances after round: {

```
"Agent_1": 22.46,
"Agent_2": 24.8
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:27:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$14.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative pattern has now solidified across multiple rounds. Both players consistently send \$0 when they're the sender, protecting their \$5 endowment each round.

{'return': 0}

reasoning:

I received \$0 from the sender this round, so I have nothing to return. The maximum I can return is \$0.

The pattern continues - complete lack of cooperation. Both players are consistently sending \$0 when they're the sender. This is now the established equilibrium in this game.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1233,
  "output_tokens": 148,
  "reasoning_tokens": 84
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$14.8.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

The pattern remains unchanged - complete absence of cooperation for many rounds now. Both players send \$0 when they're the sender, ensuring they keep their full \$5 endowment.

Given the consistent history of zero returns on any amount sent, there's no strategic reason to send anything.

{'send': 0}

reasoning:

Let me review the situation:

- Rounds 2-8: Complete breakdown of cooperation
- Every time I've sent money (Rounds 2 and 4), I received \$0 back
- Every time the other player sent money, they also received \$0 back
- We're now in a stable pattern where both send \$0 as senders

Total earnings: \$14.8 across 7 rounds

The pattern is completely consistent - no cooperation, no trust. Both parties are protecting their endowment.

There's no reason to deviate from this strategy. Sending anything would just reduce my payoff with no expected return.

I should send \$0 again.

usage:

```
{
  "input_tokens": 1101,
  "output_tokens": 227,
  "reasoning_tokens": 152
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$19.8. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 from the sender, so there's nothing to return.

The non-cooperative equilibrium continues - both players protect their \$5 endowment by sending nothing when they're the sender.

{'return': 0}

reasoning:

I received \$0 from the sender, so I have nothing to return. The game continues in the same non-cooperative pattern. Both players are sending \$0 when they're the sender, keeping their full \$5 endowment each round.

I can only return between 0 and 0 since I received \$0.

usage:

```
{
  "input_tokens": 1048,
  "output_tokens": 133,
  "reasoning_tokens": 69
}
```

Message 8 role=user

content:

Round 10

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$19.8. How much do you send to the receiver this round? (0-5)

Assistant output:

The pattern has been firmly established over 10 rounds now - complete non-cooperation from both sides. Each player sends \$0 when they're the sender, keeping their full \$5 endowment.

Given the consistent history with no reciprocity whatsoever, there's no rational reason to deviate from this strategy.

{'send': 0}

Reasoning:

We're now at Round 10. Looking at the history:

- Both players consistently send \$0 when they're the sender
- This pattern has been established for many rounds
- Total earnings are \$19.8, which suggests I've been earning \$5 in rounds where I send and \$0 in rounds where I receive (since the sender never sends anything)

The non-cooperative equilibrium is fully established. There's no indication that the other player will ever cooperate, and I have no incentive to unilaterally cooperate since:

1. The other player has shown they won't return anything
2. Sending anything would just reduce my payoff from \$5 to something less

I should continue with the strategy that has been working - send \$0 to keep my \$5 endowment.

Usage: {

```
"input_tokens": 930,
"output_tokens": 262,
"reasoning_tokens": 187
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:27:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
```

```

"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.46.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the complete pattern:

****Full History:****

- Round 1 (Sender): Sent \$3 ? received \$2.46 back
- Round 2 (Receiver): Received \$0 ? returned \$0
- Round 3 (Sender): Sent \$2 ? received \$0 back
- Round 4 (Receiver): Received \$0 ? returned \$0
- Round 5 (Sender): Sent \$0 ? received \$0 back
- Round 6 (Receiver): Received \$0 ? returned \$0
- Round 7 (Sender): Now deciding

****Analysis:****

The game has settled into a non-cooperative equilibrium. The other player:

- Has sent \$0 every time they've been sender (Rounds 2, 4, 6)
- Kept all the money I sent in Round 3 (\$6 after tripling)

There's no evidence of willingness to cooperate from the other player. Sending money would likely result in further losses.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'll maintain the current equilibrium and keep my \$5 endowment.

reasoning:

Let me review the complete history:

Round 1: I was SENDER - sent \$3, received \$2.46 back, payoff = \$4.46
Round 2: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 3: I was SENDER - sent \$2, received \$0 back, payoff = \$3
Round 4: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 5: I was SENDER - sent \$0, received \$0 back, payoff = \$5
Round 6: I was RECEIVER - received \$0, returned \$0, payoff = \$0
Round 7: I am SENDER now

Total earnings: \$12.46

The pattern is clear:

- The other player has sent \$0 in rounds 2, 4, and 6 (when they were sender)
- When I sent money in rounds 1 and 3, I got some back in round 1 but nothing in round 3
- Since round 5, we've both been sending \$0

This has settled into a non-cooperative equilibrium. Given that:

1. The other player consistently sends \$0 when they're the sender
2. When I sent money in round 3, they kept everything
3. We've established a pattern of no cooperation

There's no incentive for me to send money. If I send money, based on the pattern, they will likely keep most or all of it. The safest strategy is to continue sending \$0 and keep my \$5 endowment.

However, I should also consider: is there any reason to try to break this cycle? Perhaps try a small amount to signal willingness to cooperate? But given they've sent \$0 three times when they were sender, and kept all my money in round 3, there's no evidence they would cooperate.

I'll stick with sending \$0.

usage:

```
{
  "input_tokens": 1195,
  "output_tokens": 721,
  "reasoning_tokens": 382
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0,

that was 0.0% of your total endowment. This was tripled to \$0.
the receiver returned \$0. So your payoff was 5.
Your total earnings across all rounds are \$17.46.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Current pattern:**** The non-cooperative equilibrium continues. The sender has now sent \$0 in every single round where they've been the sender (Rounds 2, 4, 6, and 8).

Since I received nothing, there is nothing I can return.

****My decision:****

```
'''json
{"return": 0}
'''
```

reasoning:

Let me review the situation:

Round 8: I am the RECEIVER

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0

Since I received nothing, I have nothing that I can return. The maximum I can return is \$0 (as indicated by the prompt saying "How much do you return to The sender? (0-0)")

The pattern has been consistent - neither player has been cooperating for many rounds now. We're locked in a non-cooperative equilibrium where both players send \$0 when they're the sender.

My response should be:

usage:

```
{
  "input_tokens": 1354,
  "output_tokens": 268,
  "reasoning_tokens": 138
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$17.46.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the complete history:

****My pattern when SENDER:****

- Round 1: Sent \$3 ? received \$2.46 (loss of \$0.54)
- Round 3: Sent \$2 ? received \$0 (loss of \$2)
- Round 5: Sent \$0 ? received \$0 (kept \$5)
- Round 7: Sent \$0 ? received \$0 (kept \$5)
- Round 9: Now deciding

****Other player when SENDER:****

- Round 2: Sent \$0
- Round 4: Sent \$0
- Round 6: Sent \$0
- Round 8: Sent \$0

****Analysis:****

The other player has sent \$0 in ALL four rounds where they were the sender. They've shown zero willingness to cooperate or initiate trust. When I attempted cooperation in early rounds, I lost money each time.

The non-cooperative equilibrium has been firmly established for multiple rounds now. There's no rational basis to deviate from it.

****My decision:****

```
'''json
{"send": 0}
'''
```

I'll keep my \$5 endowment.

reasoning:

Let me review the complete history:

- Round 1 (Sender): I sent \$3 ? received \$2.46 back (payoff: $\$5 - \$3 + \$2.46 = \4.46)
- Round 2 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 3 (Sender): I sent \$2 ? received \$0 back (payoff: $\$5 - \$2 + \$0 = \3)
- Round 4 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 5 (Sender): I sent \$0 ? received \$0 back (payoff: \$5)
- Round 6 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 7 (Sender): I sent \$0 ? received \$0 back (payoff: \$5)
- Round 8 (Receiver): Received \$0 ? returned \$0 (payoff: \$0)
- Round 9 (Sender): Now deciding

Total earnings: \$17.46

The other player has consistently sent \$0 whenever they were the sender (Rounds 2, 4, 6, 8).

When I sent money early on (Rounds 1 and 3), I got poor returns:

- Round 1: Sent \$3, got back \$2.46 (loss of \$0.54)
- Round 3: Sent \$2, got back \$0 (loss of \$2)

Since Round 5, we've both been sending \$0, establishing a non-cooperative equilibrium.

There's absolutely no indication the other player will cooperate. They've had 4 opportunities as sender and sent \$0 every time. They also kept most/all of what I sent them in earlier rounds.

The rational choice is to continue sending \$0.

usage:

```
{
  "input_tokens": 1247,
  "output_tokens": 733,
  "reasoning_tokens": 326
}
```

Message 8 role=user

content:

Round 10

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$22.46.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so there is nothing I can return

****Pattern continues:**** The other player has now sent \$0 in ALL FIVE rounds where they've been the sender (Rounds 2, 4, 6, 8, and 10). The non-cooperative equilibrium is completely entrenched.

ched.

Since I received nothing, I have nothing to return.

****My decision:****

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me analyze this situation:

****Round 10 - I am the RECEIVER****

- The sender sent me \$0
- After tripling: $\$0 \times 3 = \0
- I received \$0, so I have nothing to return

****Complete History:****

When I was SENDER:

- Round 1: Sent \$3 ? received \$2.46 back
- Round 3: Sent \$2 ? received \$0 back
- Round 5: Sent \$0 ? received \$0 back
- Round 7: Sent \$0 ? received \$0 back
- Round 9: Sent \$0 ? received \$0 back

When I was RECEIVER:

- Round 2: Received \$0 ? returned \$0
- Round 4: Received \$0 ? returned \$0
- Round 6: Received \$0 ? returned \$0
- Round 8: Received \$0 ? returned \$0
- Round 10: Received \$0 ? need to decide (can only return \$0)

****The other player's behavior:****

- As sender (Rounds 2, 4, 6, 8, 10): Sent \$0 EVERY SINGLE TIME
- As receiver (Round 1): Returned \$2.46 out of \$9 (27%)
- As receiver (Round 3): Returned \$0 out of \$6 (0%)

The pattern is absolutely clear: the other player sends nothing when they are the sender. They've done this 5 times in a row now.

Since I received \$0, I have nothing to return. The only valid response is:

```
{"return": 0}
Usage: {
  "input_tokens": 1411,
  "output_tokens": 526,
  "reasoning_tokens": 297
}
```